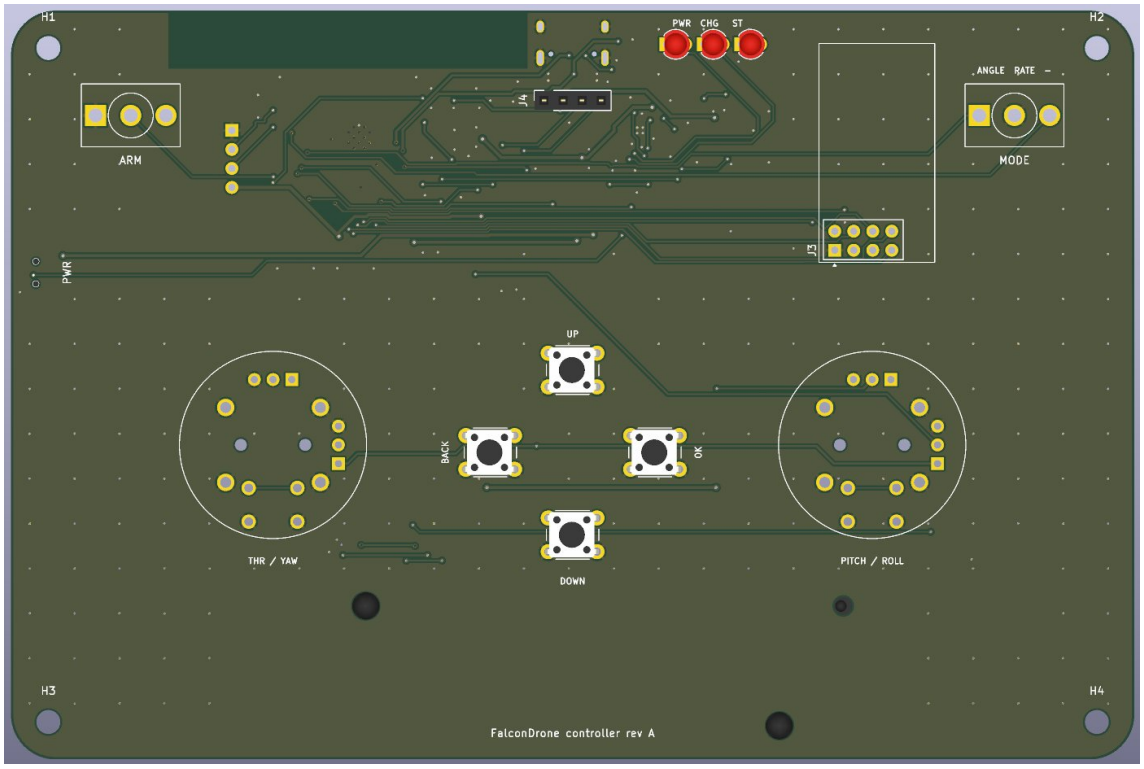
FalconDrone

Hardware design description

Handheld controller (falcon_rc) and flight controller (falcon_fc), rev A



Controller, front side

Generated 2026-10-21 by hardware/gen_design_doc.py from the KiCad 9 projects. The schematics, footprints and boards are themselves generated by the scripts described in section 4: edit the scripts, not the KiCad files.

1 System overview

FalconDrone is a quadcopter built around a custom STM32F446 flight controller. This document covers the two custom boards: the **flight controller** that flies the drone, and the **handheld controller** the pilot holds. They talk over an nRF24L01+ 2.4 GHz link: the controller sends stick and switch frames, the drone answers with telemetry in the ACK payload.

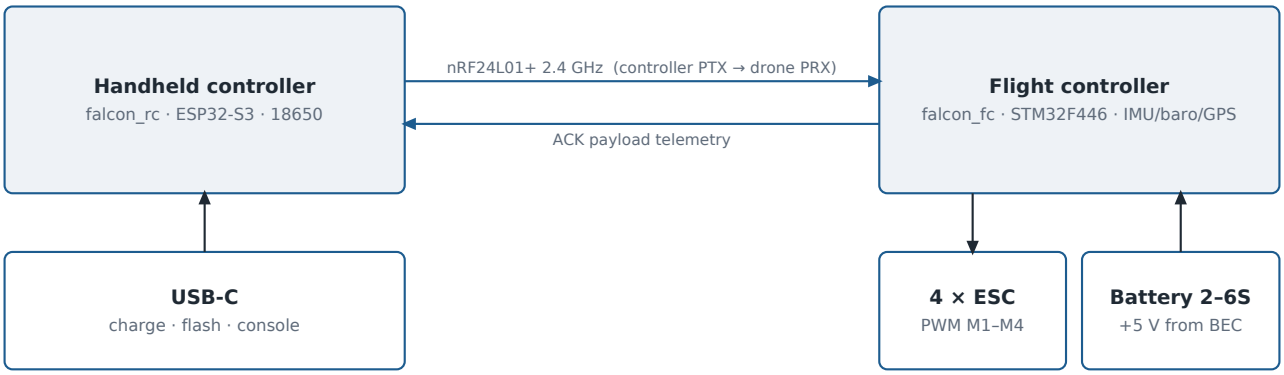


Figure 1 – system blocks

	Handheld controller	Flight controller
Project	hardware/controller/falcon_rc	hardware/flight_controller/falcon_fc
MCU	ESP32-S3-WROOM-1-N8 (Xtensa LX7, native USB)	STM32F446RET6 (Cortex-M4F, 180 MHz)
Board	150 × 100 mm, 2 layers, 1.6 mm	45 × 45 mm, 4 layers, 1.6 mm, 30.5 mm mount
Power	1 × 18650 Li-ion, USB-C charging	2–6S flight pack sense, 5 V from ESC BEC
Radio	nRF24L01+ PA/LNA module (PTX)	nRF24L01+ module (PRX)
Parts	78 footprints, 73 assembled	60 footprints
Status	routed, DRC clean, JLCPCB full-assembly files	routed, DRC clean

2 Handheld controller (falcon_rc)

A gamepad-shaped 150 × 100 mm board, held in both hands. Mode-2 layout: left stick throttle / yaw, right stick pitch / roll, a four-button menu cluster between the sticks, ARM and MODE toggles at the top corners, OLED and status LEDs top centre. The 18650 cell and every SMD part sit on the back, so the front carries only the controls and the board needs one reflow pass.

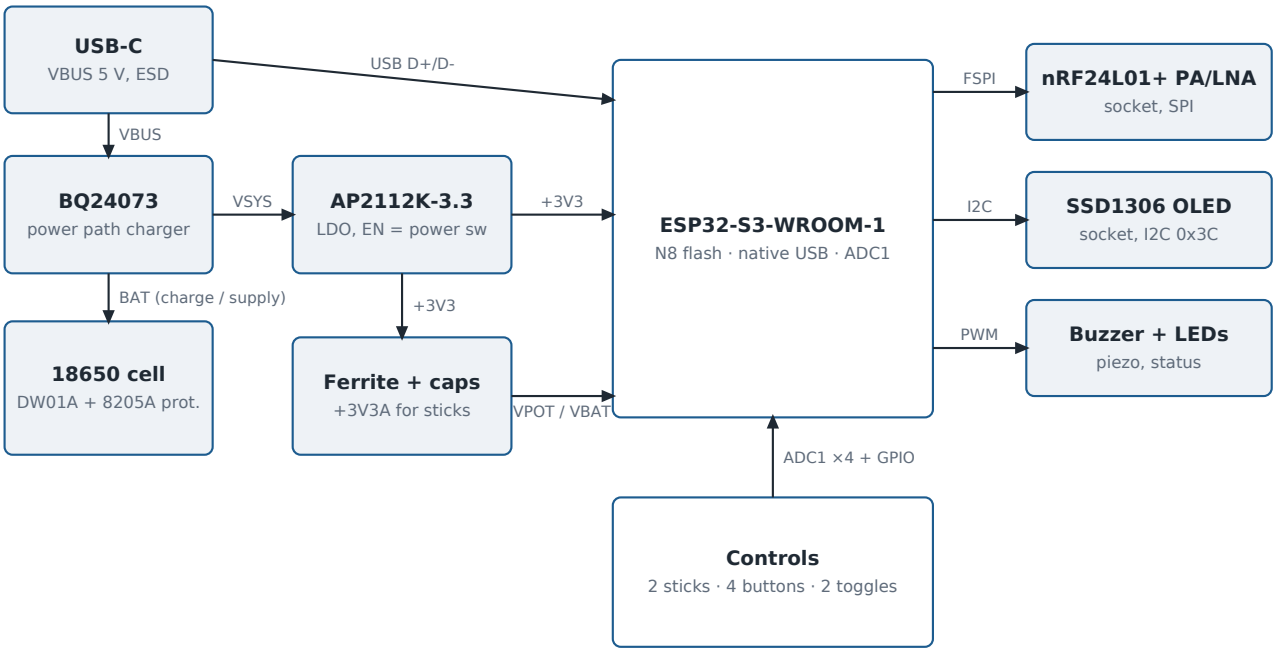


Figure 2 – controller blocks

2.1 Power path

USB-C feeds a TI **BQ24073** linear charger with power-path management: its OUT pin (VSYS) is regulated to 4.4 V while USB is present and switches to the battery otherwise, so the controller runs while charging and charges while switched off. The cell is protected in its negative lead by a **DW01A** with a dual N-MOSFET **8205A**; board GND is pack minus. An **AP2112K-3.3** LDO makes +3V3. The power slide switch drives the LDO enable pin, so it carries no load current, and the charge LEDs stay on VSYS so charging is visible with the controller off.

Setting	Part	Value	Result (TI SLUS810N)
Fast-charge current	R3 ISET	1.2 kΩ	890 AΩ / 1.2 kΩ = 0.74 A (0.66–0.81 A)
Input current limit	R4 ILIM	1.5 kΩ	1610 AΩ / 1.5 kΩ = 1.07 A (EN2 = 1, EN1 = 0)
Safety timer	R24 TMR	68 kΩ	10 × 48 s/kΩ × 68 kΩ = 9.1 h (the 5 h default could stop a 3.5 Ah cell early)
Termination	pin 15 TD	GND	termination on, at 10 % of the charge current
Battery temperature	R5 TS	10 kΩ	fixed: NTC check off (no NTC in the holder)
Input OVP	—	6.6 V	USB only; suspends above, survives 26 V
Thermal	EP + 4 vias	—	≈ 1 W at 3.6 V cell; TJ(REG) 125 °C throttles beyond

Why the BQ24073 and not the BQ24074 of the first draft: JLCPCB stocked one BQ24074. The '73 has the same 4.4 V OUT and the same ISET/ILIM/TMR constants; it differs in OVP (6.6 V instead of 10.5 V, irrelevant on USB) and in pin 15, TD instead of ITERM, tied low for the same 10 % termination.

2.2 Protection, supply budget and sensing

Block	Design	Notes
Cell protection	DW01A + 8205A (TSSOP-8), R8 100 Ω / C4 100 nF supply filter, R9 1 kΩ CS	over-charge 4.3 V, over-discharge ~2.4–2.5 V, short circuit via the FETs' $R_{DS(on)}$

Block	Design	Notes
3.3 V rail	AP2112K-3.3, 600 mA, C5/C6 10 µF	peak load ≈ 350 mA (ESP32-S3 RF + nRF24 PA bursts), ≈ 0.17 W typical in the LDO. Drops out below ≈ 3.5 V cell: firmware warns at 3.6 V, shuts down at 3.4 V
Analog rail	FB1 600 Ω ferrite, C8 10 µF + C9 100 nF → +3V3A	feeds the four stick pots; VPOT_SENSE reads it so stick values are ratiometric
Sense dividers	VBAT 220k/220k, VPOT 100k/100k, VBUS 100k/150k	1:2, 1:2 and 0.6 into ADC1; 100 nF / 10 nF at the pins; µA drain when off
USB	HRO TYPE-C-31-M-12, 5.1 kΩ CC pull-downs, USBLC6-2SC6	native ESP32-S3 USB: flashing, console and charging on one port

2.3 ESP32-S3 pin map

ADC inputs use ADC1 only (ADC2 conflicts with Wi-Fi). IO35–37 stay free so an octal-PSRAM module variant also fits; the 1.8 V “V” variants are excluded (their IO47/48 run at 1.8 V). Buttons and switches pull to GND with the internal pull-ups.

Function	GPIO	Function	GPIO
STICK_LX (yaw)	`IO1`	OLED SDA / SCL	`IO8 / IO9`
STICK_LY (throttle)	`IO2`	nRF24 CSN / MOSI / SCK / MISO	`IO10 / 11 / 12 / 13`
STICK_RX (roll)	`IO4`	nRF24 CE / IRQ	`IO14 / IO21`
STICK_RY (pitch)	`IO5`	BTN_L / BTN_R (stick press)	`IO15 / IO16`
VBAT_SENSE	`IO6`	BTN UP / DOWN / OK / BACK	`IO17 / 18 / 38 / 39`
VPOT_SENSE	`IO7`	SW_ARM	`IO40`
VBUS_DET	`IO3`	SW_MODE_A / B (centre = neither)	`IO41 / IO42`
USB D- / D+	`IO19 / IO20`	BUZZER / LED_STATUS	`IO47 / IO48`
BOOT button	`IO0`	Debug UART (J2, not fitted)	`TXD0 / RXD0`

2.4 Radio, display and controls

- **Radio:** nRF24L01+ PA/LNA module in a 2 × 4 socket (J3) on FSPI IO-MUX pins, 22 µF + 100 nF at the socket for the ≈ 115 mA TX bursts. Its SMA antenna points off the top edge, at the opposite end from the ESP32's PCB antenna, whose keep-out is clear of copper on both layers.
- **Display:** 0.96" SSD1306 module in a 1 × 4 socket (J4), pin order GND VCC SCL SDA, 4.7 kΩ I2C pull-ups. Some modules swap GND/VCC: check before plugging in.
- **Sticks:** bare Alps RKJXV1224005 (two 10 kΩ pots + push switch) on a custom footprint drawn from Alps drawing KJXV14005. Frame tabs soldered to GND; C18–C21 100 nF at the ADC pins. The left stick's Y spring can be removed for a non-centring throttle.
- **Switches:** C&K 7101 (ARM, ON-ON) and 7103 (MODE, ON-OFF-ON: ANGLE / RATE / spare) with PC pins on 4.70 mm, 1.85 mm holes per C&K's layout. MTS-102/103 clones fit the same footprint. Four 6 × 6 mm tact switches form the menu cluster.
- **Buzzer:** CUI CPT-9019S piezo (12 nF, 4 kHz) switched by an MMBT3904; R25 1 kΩ across it discharges the piezo while the transistor is off.
- **LEDs:** 3 mm through-hole: green PWR (PGOOD), red CHG (charging), yellow status (IO48). Yellow because a blue LED's ≈ 3.4 V forward voltage would not light from a 3.3 V pin.
- **Reset / boot:** SMD RESET and BOOT buttons on the back, R17/C15 delay EN until 3V3 is up. Native USB normally auto-resets into download mode.

2.5 PCB

Item	Value
Outline	150 × 100 mm, 6 mm corner radius, 4 × M3 holes 5 mm in from the corners
Stack-up	2 layers, 1.6 mm FR-4, 1 oz; GND pours on both layers
Rules	JLCPCB standard: 0.15 mm clearance, 0.3 mm min drill, 0.6 mm via
Placement	front: sticks, toggles, tact switches, LEDs, nRF24 and OLED sockets; back: every SMD part, 18650 holder, USB-C, power switch
Routing	890 track segments, 389 vias (incl. GND fan-out, charger thermal vias and the stitching grid)
Checks	ERC 0 violations · 0 unconnected · 0 schematic-parity issues · DRC remaining: 1 lib footprint mismatch, 3 silk edge clearance, 7 starved thermal

Remaining DRC entries are expected: the ESP32 footprint differs from the library because its 0.2 mm GND-pad vias were widened to JLCPCB's 0.3 mm minimum; the power switch's lever silk overhangs the edge by design; and a few GND pads join the pour with one thermal spoke but also have their own via.

Net classes

Class	Track	Clearance	Nets
Battery	0.4 mm	0.15 mm	VBUS, VSYS, BATT_P, BATT_N (≈ 7 °C rise at 1 A)
Power	0.4 mm	0.15 mm	+3V3, GND
Analog	0.25 mm	0.2 mm	+3V3A, STICK_*, VBAT_SENSE, VPOT_SENSE
USB	0.3 mm	0.15 mm	USB_DP / USB_DN, short and coupled
Default	0.2 mm	0.15 mm	everything else

0.4 mm, not wider, for the battery path: wider tracks could not enter the charger's 0.5 mm-pitch QFN pads, and 0.4 mm on 1 oz copper carries 1 A at ≈ 7 °C rise (IPC-2221).

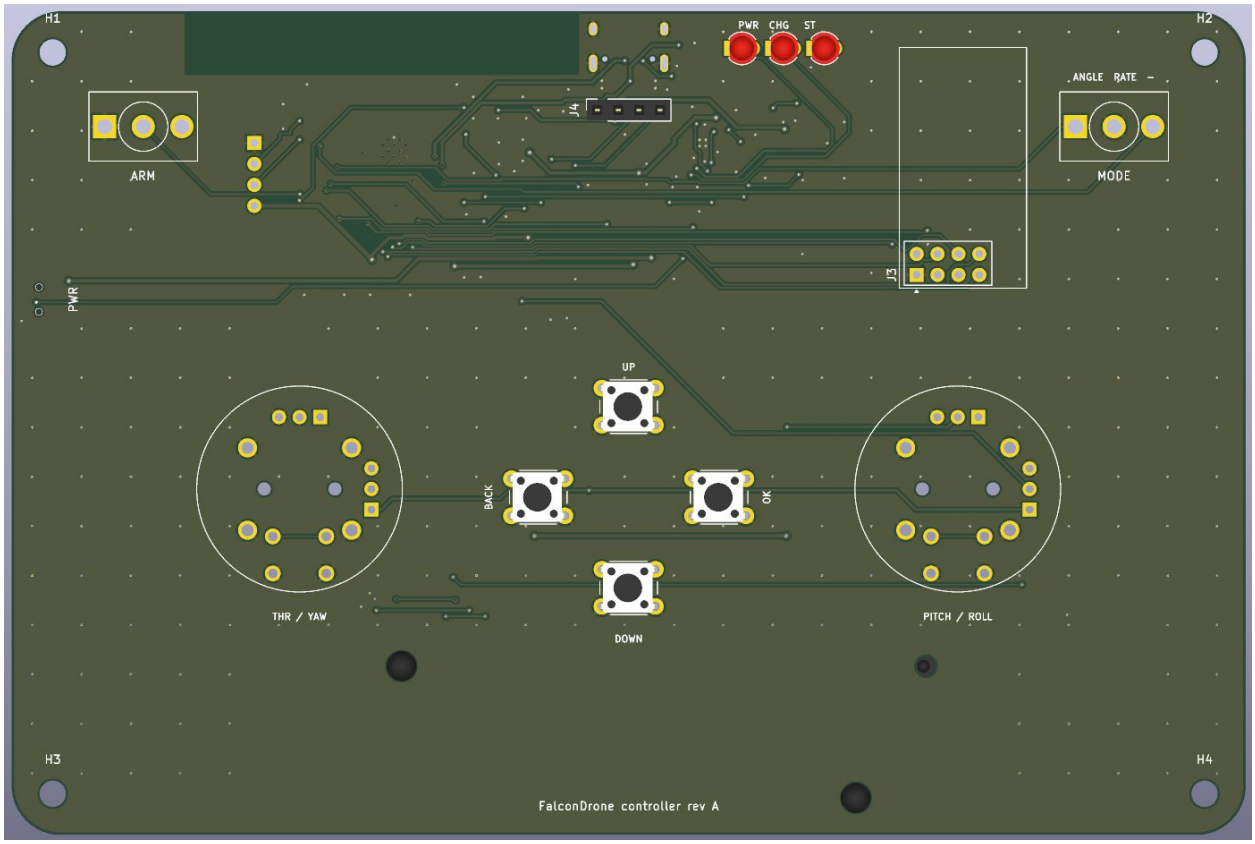


Figure 3 – front: controls only (through-hole)

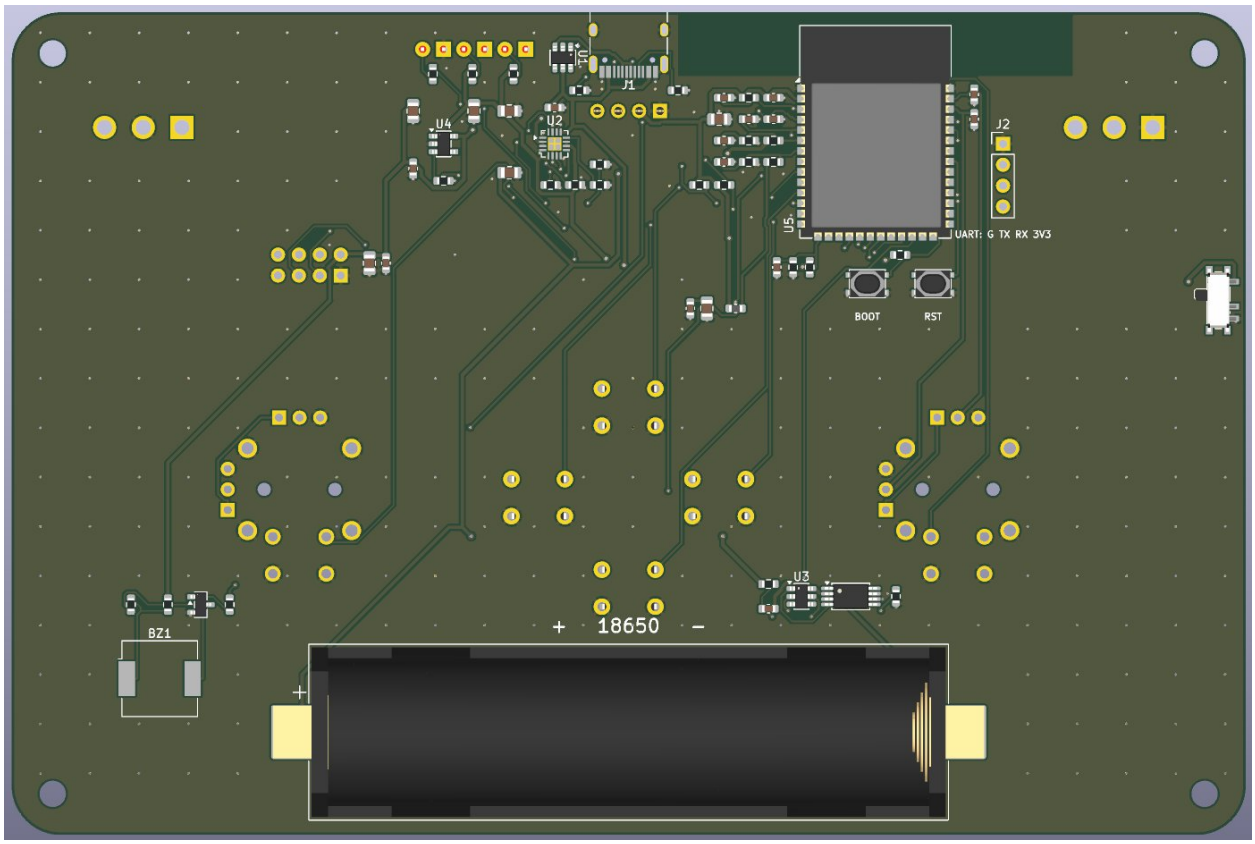


Figure 4 – back: all SMD parts, 18650 holder, USB-C

2.6 Manufacturing (JLCPCB, full assembly)

`controller/gen_fab.py` writes the order files to `controller/build/jlcpcb/`: the Gerber + drill zip, the BOM with an LCSC number per line and the CPL (placement) file. SMD parts are reflowed on the back; the through-hole controls on the front are soldered by JLCPCB's THT service. Only J2 (debug UART header) is not fitted, plus the four mounting holes, which are not parts. `gen_fab.py --stock` re-checks live stock.

Qty	Designators	Part	LCSC	Side
1	BT1	Keystone 1042	C5249310	back (reflow)
1	BZ1	CUI CPT-9019S-SMT-TR	C95163	back (reflow)
2	C1, C15	1u	C15849	back (reflow)
5	C2, C5, C6, C8, C13	10u	C15850	back (reflow)
1	C3	4.7u	C1779	back (reflow)
11	C4, C7, C9, C10, C11, C14, C17, C18, C19, C20, C21	100n	C14663	back (reflow)
1	C12	10n	C57112	back (reflow)
1	C16	22u	C45783	back (reflow)
1	D1	Everlight 204-10SYGD/S530-E3 green	C85161	front (THT)
1	D2	Everlight 204-10SDRD/S530-A3-L red	C84774	front (THT)
1	D4	Everlight 204-10UYD/S530-A3-L yellow	C85160	front (THT)
1	FB1	Sunlord GZ1608D601TF	C1002	back (reflow)
1	J1	HRO TYPE-C-31-M-12	C165948	back (reflow)
1	J3	CJT A2541HWV-2x4P (nRF24 socket)	C239344	front (THT)
1	J4	BOOMELE 2.54-1*4P female (OLED socket)	C2718488	front (THT)
2	JS1, JS2	Alps RKJXV1224005	C146170	front (THT)
1	Q1	HXY 8205A (TSSOP-8, FS8205A pinout)	C5148694	back (reflow)
1	Q2	MMBT3904	C20526	back (reflow)
2	R1, R2	5.1k	C23186	back (reflow)
1	R3	1.2k	C22765	back (reflow)
1	R4	1.5k	C22843	back (reflow)
3	R5, R17, R18	10k	C25804	back (reflow)
6	R6, R7, R9, R21, R23, R25	1k	C21190	back (reflow)
1	R8	100	C22775	back (reflow)
4	R10, R11, R12, R15	100k	C25803	back (reflow)
2	R13, R14	220k	C22961	back (reflow)
1	R16	150k	C22807	back (reflow)
2	R19, R20	4.7k	C23162	back (reflow)
1	R22	22	C23345	back (reflow)
1	R24	68k	C23231	back (reflow)
1	SW1	C&K PCM12SMTR	C221841	back (reflow)
2	SW2, SW3	C&K PTS810SJG250SMTRLFS	C221895	back (reflow)
4	SW4, SW5, SW6, SW7	HRO K2-1102DP-C4SW-04	C110153	front (THT)
1	SW8	C&K 7101SYCQE	C221549	front (THT)
1	SW9	C&K 7103SYCQE (Global Sourcing)	Global Sourcing	front (THT)
1	U1	ST USBLC6-2SC6	C7519	back (reflow)
1	U2	TI BQ24073RGTR	C15220	back (reflow)
1	U3	PUOLOP DW01A	C351410	back (reflow)
1	U4	Diodes AP2112K-3.3TRG1	C51118	back (reflow)
1	U5	Espressif ESP32-S3-WROOM-1-N8	C2913198	back (reflow)

Ordering notes

- All resistors and MLCCs are JLCPCB Basic parts (no per-part setup fee).
- SW9 (C&K 7103SYCQE) has no LCSC stock: choose it under Global Sourcing on the BOM page, or consign it. SW8 (7101SYCQE) had only 2 in stock on 2026-10-21.
- The CPL uses the kicad-jlcpcb-tools conventions (pad-centre position, mirrored bottom angles) and JLCKicadTools' rotation corrections. Check U2, U5, J1 and Q1 in JLCPCB's placement preview before paying.
- Low stock to watch: CPT-9019S buzzer, Keystone 1042 holder.

3 Flight controller (falcon_fc)

A 45 × 45 mm, 4-layer board on the standard 30.5 mm stack pattern (4 mm holes for M3 rubber grommets). It replaces the Nucleo-64 + breakout bench setup: MCU, IMU, barometer, GPS, radio socket, ESC and AUX connectors on one board, with the pin assignment kept to the existing CubeMX project (falconDrone.ioc) wherever possible.

Block	Part	Notes
MCU	STM32F446RET6 (LQFP-64)	one 100 nF per VDD pin, low-ESR ceramic on VCAP_1, SWD + SWO on a JST-GH that matches Nucleo CN4
IMU	TDK ICM-42688-P	SPI3 ≤ 24 MHz, INT1 to PB5, near the centre of mass
Barometer	MS5611-01BA03	I2C1 at 0x77, away from the regulators, cover the port with open-cell foam
GPS	u-blox SAM-M10Q	USART6, own LP5907 LDO, solid GND under the patch antenna
Radio	nRF24L01+ module socket	SPI2, module overhangs the south edge
Power in	JST-GH: VBAT, GND, +5 V	AO3401A reverse-polarity P-FET, SMAJ5.0A TVS
3.3 V	AP7361C-33E (SOT-223)	main rail, ≈ 0.3 W worst case into a GND pour
Sensor 3.3 V	LP5907MFX-3.3	quiet rail for IMU + baro (+3V3_SENS)
Battery sense	10k / 1k divider	6S 25.2 V → 2.29 V on PA4
ESD	3 × SP0504BAHT	at the ESC, AUX and telemetry connectors

3.1 Pin map

Function	Pins	Function	Pins
ESC M1–M4	PA0 PA1 (TIM2) · PA6 PA7 (TIM3)	IMU SPI3	PC10 SCK · PC11 MISO · PC12 MOSI
nRF24 SPI2	PB13 SCK · PB14 MISO · PB15 MOSI	IMU CS / INT1	PA15 / PB5
nRF24 CSN / CE / IRQ	PB12 / PB1 / PC4	Baro I2C1	PB6 SCL · PB7 SDA
Telemetry USART2	PA2 TX · PA3 RX	GPS USART6	PC6 TX · PC7 RX
AUX USART1 + I2C3	PA9 PA10 · PA8 PC9	VBAT_SENSE	PA4 (ADC)
SWD	PA13 · PA14 · PB3 SWO	LED / CAL button	PA5 / PC13

New pins, not yet in the .ioc: SPI3 IMU, I2C1 baro, PA4 VBAT_SENSE and PC4 NRF_IRQ. The RC link on this board is the nRF24 only (build with FALCON_RC_RADIO=NRF24).

3.2 PCB

Item	Value
Stack-up	F.Cu signals · In1 GND plane · In2 +3V3 plane · B.Cu signals, 1.6 mm
Placement	top: GPS centre, power entry north, ESC header east, AUX west, nRF24 south; bottom: MCU under the GPS (shielded by In1), IMU near the centre, baro apart
Routing	699 track segments, 132 vias; maze-routed IMU bus and local links first, then Freerouting in stages
Checks	ERC 0 violations · 0 unconnected · 0 schematic-parity issues · DRC remaining: 1 lib footprint mismatch (the nRF24 footprint was edited on purpose)

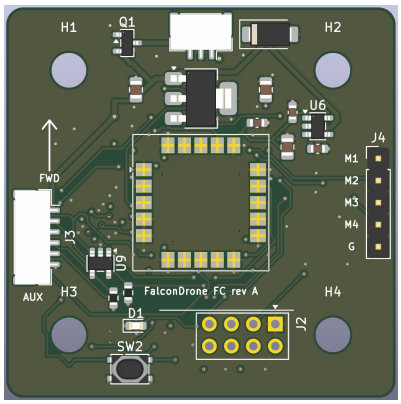


Figure 5 – FC top

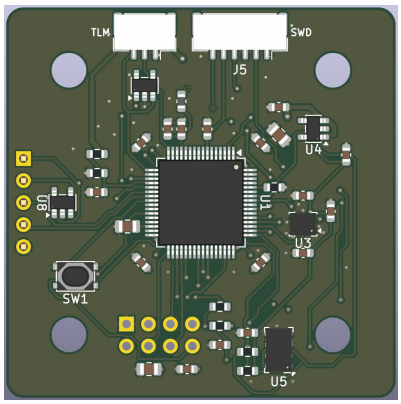


Figure 6 – FC bottom

4 Design pipeline

Everything is generated from Python, so every change is a reviewable diff and the boards can be rebuilt from scratch. KiCad is used to view, check and export. Hand edits to the .kicad_* files are lost the next time a script runs.

Step	Script	Output
Shared helpers	`hardware/schgen.py`	label-only schematic writer, symbol library handling, R/C/header helpers
Schematic	`gen_schematic.py`	*.kicad_sch + project symbols; then kicad-cli sch erc
Footprints	`controller/gen_footprints.py`	falcon_rc.pretty (Alps stick, 4.7 mm toggle)
Placement	`gen_pcb.py [--force]`	outline, rules, net classes, placed parts, pours
Routing	`route_pcb.py`	GND fan-out → Freerouting → maze finisher → pours → stitching vias → GND-island join
Order files	`controller/gen_fab.py`, `flight_controller/export_fab.py`	Gerbers + drill zip, BOM, CPL in build/jlcpcb/
This document	`hardware/gen_design_doc.py`	FalconDrone_hardware_design.pdf

route_pcb.py needs Java 25+ and the Freerouting 2.x jar: `FREEROUTING_JAR=… JAVA=… python3 route_pcb.py --force`. build/ directories are git-ignored.

5 Verification status and open items

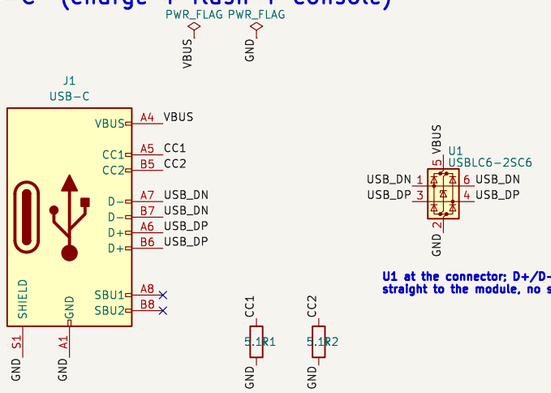
Verified

- Charger, protection, LDO, ESD, buzzer driver and FET pinouts checked against the datasheets (TI SLUS810N, Fortune/HXY 8205A, CUI CPT-9019S, C&K 7000 series).
- Pin-by-pin netlist audit of the controller; ERC clean on both schematics; both boards DRC clean with schematic parity.
- Every controller part matched to a stocked JLCPCB part (one Global Sourcing).

Open before / after the first build

- Controller firmware: retarget from ESP32 to ESP32-S3 with a new board.h for the pin map in 2.3.
- Print the Alps stick footprint 1:1 and test-fit a real stick (the switch-row position was measured off the Alps drawing, not dimensioned).
- Check part rotations in JLCPCB's placement preview (U2, U5, J1, Q1).
- 1.07 A input limit can overload a USB-2 port that only allows 500 mA; phone and USB-C chargers are fine.
- Check the OLED module's pin order (GND/VCC) before plugging it in.
- Flight controller: add the new pins (SPI3, I2C1, PA4, PC4) to falconDrone.ioc.

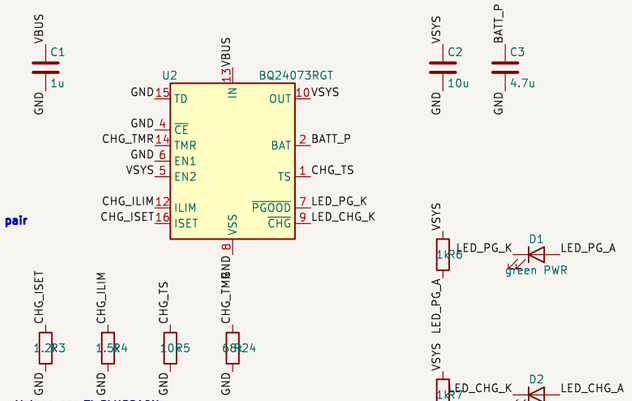
USB-C (charge + flash + console)



U1 at the connector; D+/D- 90 ohm pair straight to the module, no stubs

R1/R2: sink Rd, the source then gives 5 V (no Rd = a C-to-C charger stays off)

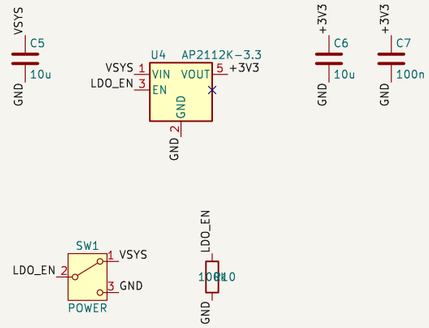
CHARGER BQ24073 (power path)



Values per TI SLUS810N:
R3 ISET 1.2k: charge 890 A-ohm / 1.2k = 0.74 A (0.66-0.81)
R4 ILIM 1.5k: input limit 1610 / 1.5k = 1.07 A (EN2=1 EN1=0)
R24 TMR 68k: fast-charge timer 10 x 48 s/k x 68k = 9.1 h (6.8 h min; the 5 h default can stop a 3.5 Ah cell early)
R5 TS: fixed 10k, temperature check off (no NTC in the holder). Swap for a 10k NTC on the cell to enable it.
TD low = termination on at 10 %. Thermal pad to GND with vias: ~1 W at 3.6 V cell, TJ(REG) 125 C throttles beyond.

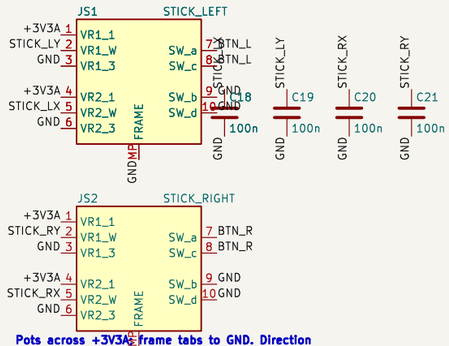
LEDs from VSYS: they work with the power switch off

3V3 AP2112K + POWER SWITCH



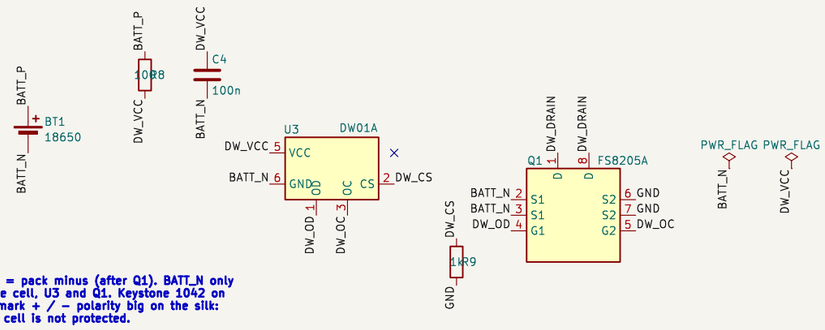
SW1 slide switch on the LDO enable, no load current through it

STICKS Alps RKJXV1224005



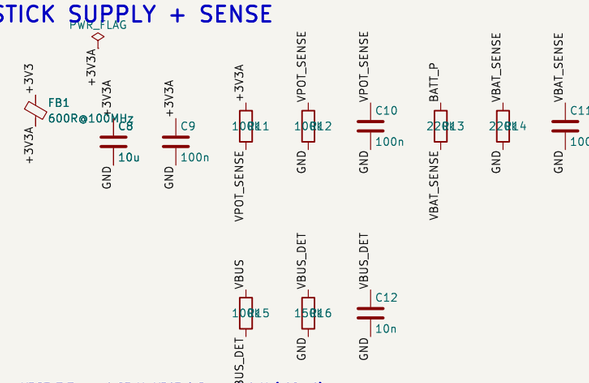
Pots across +3V3A and frame tabs to GND. Direction per axis is a firmware setting. C18-C21 at the ESP32 pins: charge reservoir for the ADC sample cap. Mode 2: left Y throttle, left X yaw, right X roll, right Y pitch. Left Y self-centres; remove its spring for a throttle that stays put.

18650 + PROTECTION DW01A / FS8205A



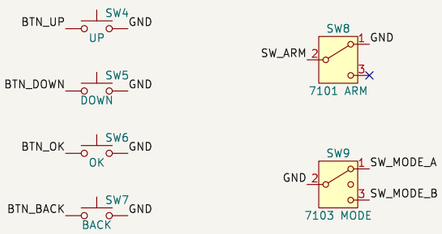
Board GND = pack minus (after Q1). BATT_N only between the cell, U3 and Q1. Keystone 1042 on the back; mark + / - polarity big on the silk: a reversed cell is not protected.

STICK SUPPLY + SENSE



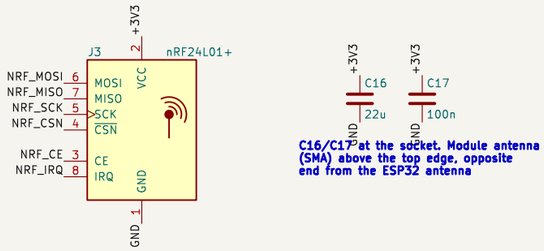
VPOT 3.3 -> 1.65 V, VBAT 4.2 -> 2.1 V (~10 uA). VBUS 5 -> 3.0 V. Read with 12 dB attenuation.

BUTTONS + SWITCHES



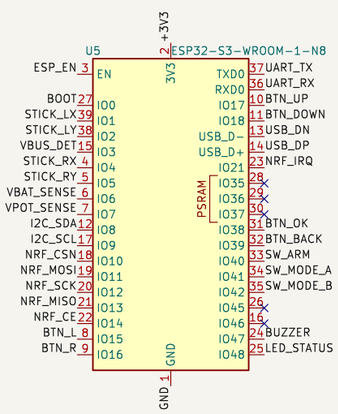
SW4-7 menu / trims. SW8 ARM toggle (ON-ON). SW9 MODE toggle (ON-OFF-ON: ANGLE / RATE / spare). Top-face toggles, PC pins at 4.7 mm: C&K 7101SYCQE / 7103SYCQE (MTS-102/103 fit too).

RADIO nRF24L01+ (PA+LNA) module



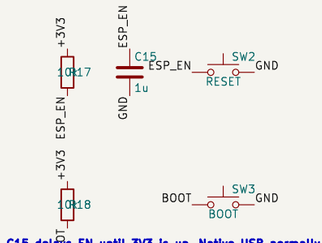
C16/C17 at the socket. Module antenna (SMA) above the top edge, opposite end from the ESP32 antenna

MCU ESP32-S3-WROOM-1





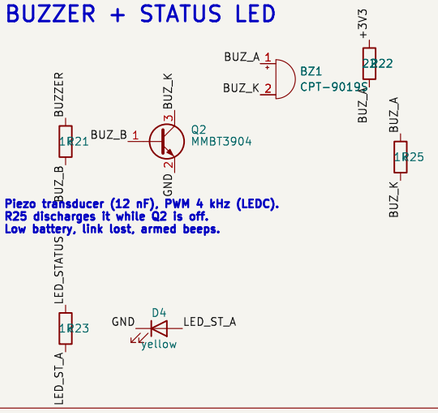
C13/C14 at module pin 2. Antenna end over the board edge, no copper under it (keepout).



C15 delays EN until 3V3 is up. Native USB normally auto-resets into download; hold BOOT + tap RESET if the firmware has broken USB.



BUZZER + STATUS LED



Piezo transducer (12 nF), PWM 4 kHz (LEDC). R25 discharges it while Q2 is off. Low battery, link lost, armed beeps.

OLED SSD1306 128x64 module (I2C 0x3C)



Pin order GND VCC SCL SDA (most 0.96" modules; some swap GND/VCC, check before soldering)

Generated by gen_schematic.py – edit the script, not this file

FalconDrone

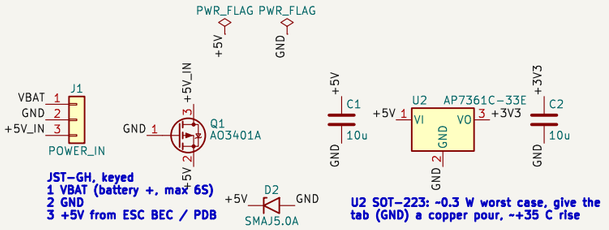
Sheet: /

File: falcon_rc.kicad_sch

Title: FalconDrone controller

Size: A3	Date:	**Rev: A**
KiCad E.D.A. 9.0.9		Id: 1/1

POWER

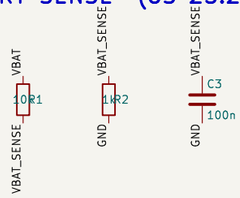


JST-GH, keyed
1 VBAT (battery +, max 6S)
2 GND
3 +5V from ESC BEC / PDB

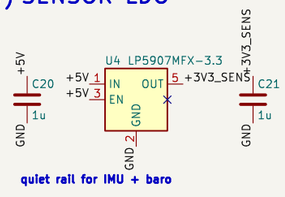
U2 SOT-223: ~0.3 W worst case, give the tab (GND) a copper pour, ~+35 C rise

Q1 reverse-polarity FET, D2 5 V TVS

BATTERY SENSE (6S 25.2 V -> 2.29 V)

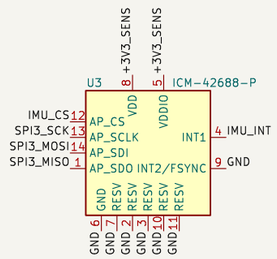


SENSOR LDO

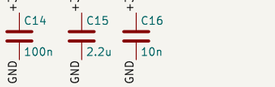


quiet rail for IMU + baro

IMU ICM-42688-P (SPI3, mode 0 or 3, <= 24 MHz)

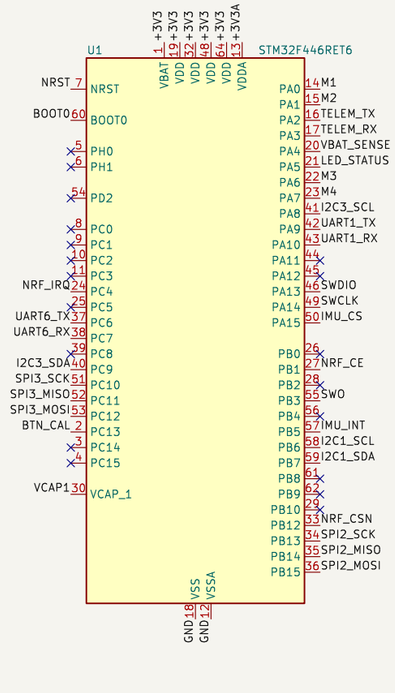


No CS pull-up needed: PA15 comes out of reset as JTDI with its internal pull-up, so CS stays high while the MCU is in reset and the IMU never latches into I2C mode

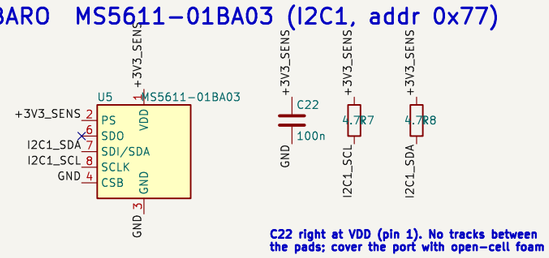


C14/C15 at VDD (pin 8), C16 at VDDIO (pin 5)

MCU STM32F446RET6

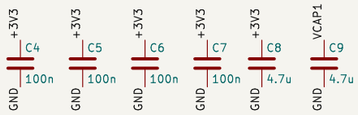


BARO MS5611-01BA03 (I2C1, addr 0x77)

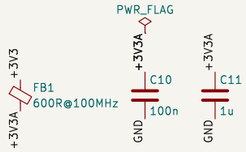


C22 right at VDD (pin 1). No tracks between the pads; cover the port with open-cell foam

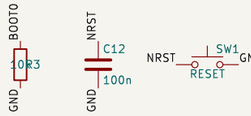
MCU DECOUPLING



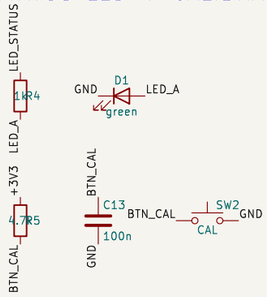
one 100n per VDD pin, close to the pin; C9 on VCAP_1: low-ESR ceramic



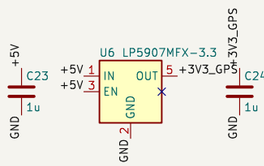
BOOT / RESET

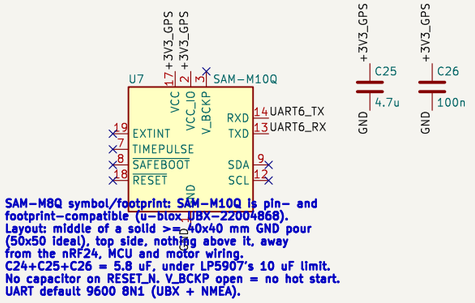


STATUS LED + CALIBRATE BUTTON



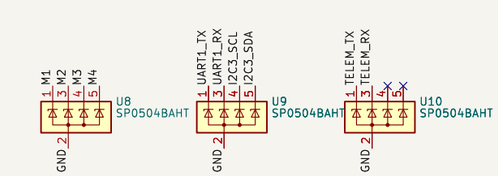
GPS SAM-M10Q (USART6)





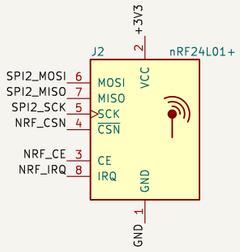
SAM-M8Q symbol/footprint: SAM-M10Q is pin- and footprint-compatible (u-blox UBX-22004868).
Layout: middle of a solid >= 40x40 mm GND pour (50x50 ideal), top side, nothing above it, away from the nRF24, MCU and motor wiring.
C24+C25+C26 = 5.8 uF, under LP5907's 10 uF limit.
No capacitor on RESET_N. V_BCKP open = no hot start.
UART default 9600 8N1 (UBX + NMEA).

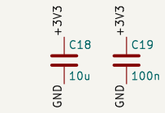
ESD (at the connectors)



U8 ESC (J4), U9 AUX (J3), U10 telemetry (J6)

RADIO nRF24L01+ module (SPI2)





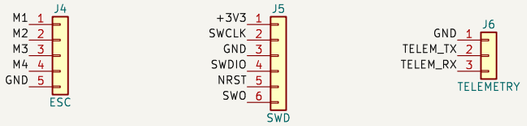
C18/C19 at the module pins: PA+LNA modules draw ~115 mA bursts

AUX (USART1 + I2C3)



JST-GH, Pixhawk GPS pinout:
1 5V 2 TX 3 RX 4 SCL 5 SDA 6 GND
I2C3 is separate from the baro bus, so 5 V pull-ups on an external module only reach 5V-tolerant pins

CONNECTORS



JST-GH, pinout = Nucleo CN4:
GH-to-Dupont cable to its ST-LINK

Generated by gen_schematic.py — edit the script, not this file

FalconDrone

Sheet: /
File: falcon_fc.kicad_sch

Title: FalconDrone flight controller

Size: A3	Date:	**Rev: A**
KiCad E.D.A. 9.0.9		Id: 1/1